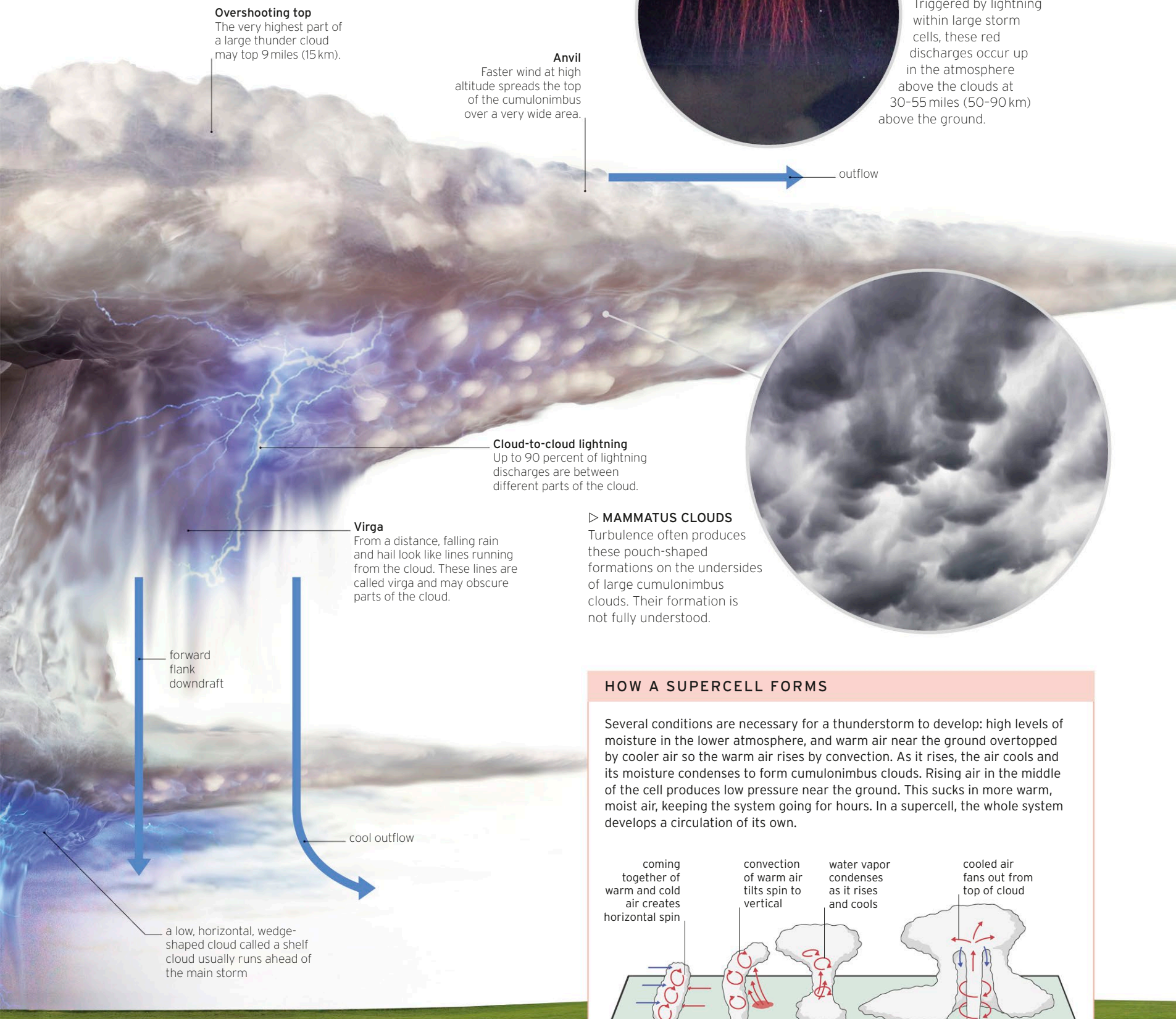


At any one time about **2,000 thunderstorms** are **taking place**, mostly in wet tropical regions



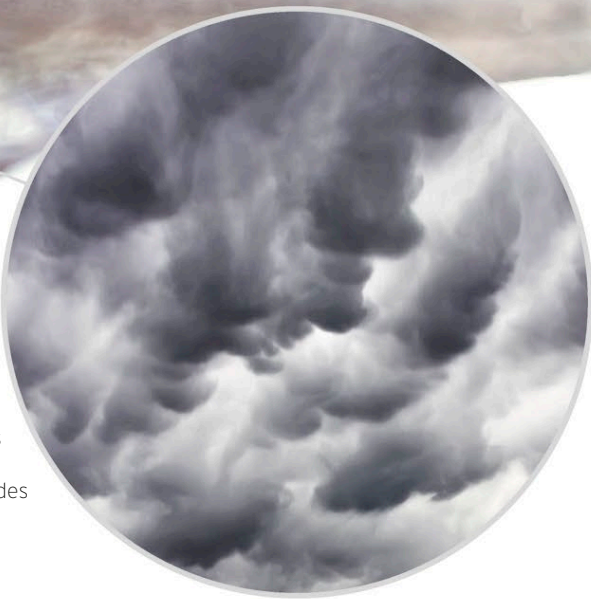
Overshooting top
The very highest part of a large thunder cloud may top 9 miles (15 km).

Anvil
Faster wind at high altitude spreads the top of the cumulonimbus over a very wide area.



◁ **RED SPRITES**
Triggered by lightning within large storm cells, these red discharges occur up in the atmosphere above the clouds at 30-55 miles (50-90 km) above the ground.

Cloud-to-cloud lightning
Up to 90 percent of lightning discharges are between different parts of the cloud.



▷ **MAMMATUS CLOUDS**
Turbulence often produces these pouch-shaped formations on the undersides of large cumulonimbus clouds. Their formation is not fully understood.

Virga
From a distance, falling rain and hail look like lines running from the cloud. These lines are called virga and may obscure parts of the cloud.

forward flank downdraft

cool outflow

a low, horizontal, wedge-shaped cloud called a shelf cloud usually runs ahead of the main storm

HOW A SUPERCELL FORMS

Several conditions are necessary for a thunderstorm to develop: high levels of moisture in the lower atmosphere, and warm air near the ground overtopped by cooler air so the warm air rises by convection. As it rises, the air cools and its moisture condenses to form cumulonimbus clouds. Rising air in the middle of the cell produces low pressure near the ground. This sucks in more warm, moist air, keeping the system going for hours. In a supercell, the whole system develops a circulation of its own.

